Adaptive Heuristic Portfolio TSP Report

Overview

- Condition count: 8.
- Best held-out TSPLIB gap: phase8_best_single_fixed_heuristic.
- Best combined transfer gap: phase8_oracle_selector.
- Lowest selector regret: phase8_best_single_fixed_heuristic.
- Pareto-efficient conditions: phase8_full_solver_evolution, phase8_llm_evolved_adaptive_controller, phase8_llm_evolved_controller_diversity_failure_replay, phase8_llm_static_selector, phase8_random_portfolio, phase8_supervised_ml_selector.

Run Metadata

- run_name: run_20260520_192410_g
- started_at_local: 2026-05-20 19:24:10
- finished_at_local: 2026-05-20 19:52:00
- duration_hhmm: 00:28
- duration_seconds: 1670.934
- seed_offset: 6000
- replicate_label: g
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Runtime (ms)	Runtime-Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_best_single_fixed_heuristic	best_fixed	none	0.07226	0.0	1475.563671	0.07226	0.007284	0.031222	0.0	0.0	False
phase8_random_portfolio	random_portfolio	none	0.171446	0.099186	374.948486	0.171446	0.083075	0.115633	0.0	0.0	True
phase8_oracle_selector	oracle_selector	none	0.07226	0.0	1531.473886	0.074998	8.9e-05	0.026678	0.0	0.0	False

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Runtime (ms)	Runtime-Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_supervised_ml_selector	supervised_selector	none	0.07226	0.0	1452.551343	0.07226	0.007284	0.031222	0.0	0.0	True
phase8_llm_static_selector	static_selector	none	0.229271	0.157011	270.031986	0.229271	0.043202	0.111754	0.0	0.62	True
phase8_llm_evolved_adaptive_controller	adaptive_controller	none	0.214311	0.142051	333.649571	0.214311	0.045346	0.107596	0.36801	0.62	True
phase8_llm_evolved_controller_diversity_failure_replay	adaptive_controller	diversity_failure	0.201354	0.129094	319.0006	0.201354	0.024368	0.089574	0.702736	1.7375	True
phase8_full_solver_evolution	full_solver	none	0.125551	0.053291	440.231957	0.125551	0.001014	0.046896	0.755521	0.76	True

Condition Notes

phase8_best_single_fixed_heuristic

- Execution mode best_fixed with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1475.563671 ms, runtime inflation 0.0, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.
- Fixed heuristic choice: farthest_insertion_two_opt.

phase8_random_portfolio

- Execution mode random_portfolio with replay none.
- Final held-out gap 0.171446, selector regret 0.099186, runtime-adjusted gap 0.171446.
- Family transfer 0.083075 and combined transfer 0.115633.
- Runtime 374.948486 ms, runtime inflation -0.745895, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

phase8_oracle_selector

- Execution mode oracle_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.074998.

- Family transfer 8.9e-05 and combined transfer 0.026678.
- Runtime 1531.473886 ms, runtime inflation 0.037891, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_supervised_ml_selector

- Execution mode supervised_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1452.551343 ms, runtime inflation -0.015596, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

phase8_llm_static_selector

- Execution mode static_selector with replay none.
- Final held-out gap 0.229271, selector regret 0.157011, runtime-adjusted gap 0.229271.
- Family transfer 0.043202 and combined transfer 0.111754.
- Runtime 270.031986 ms, runtime inflation -0.816997, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: True.
- Controller signature: nearest_neighbor_multistart:farthest_insertion_two_opt:farthest_insertion_two_opt:annealed_multi_start:stag3:cand2:restart1.
- Controller rule summary: {"acceptance_bias": 0.005, "candidate_limit_offset": 2, "clustered_heuristic": "farthest_insertion_two_opt", "corridor_heuristic": "farthest_insertion_two_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive selector over frozen portfolio. Biases toward farthest-insertion + 2-opt as default; swaps to NN-trap-safe candidate pruning and occasional annealing/multi-start for instability descriptors.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "farthest_insertion_two_opt", "low_improvement_threshold": 0.08, "name": "det_static_selector_farthest_insertion_priority", "nearest_neighbor_trap_heuristic": "candidate_pruned_two_opt", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1, "stagnation_switch_heuristic": "annealed_multi_start", "stagnation_threshold": 3, "temperature_scale": 1.0, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}

phase8_llm_evolved_adaptive_controller

- Execution mode adaptive_controller with replay none.
- Final held-out gap 0.214311, selector regret 0.142051, runtime-adjusted gap 0.214311.
- Family transfer 0.045346 and combined transfer 0.107596.
- Runtime 333.649571 ms, runtime inflation -0.773883, mean novelty 0.36801, and complexity 0.62.
- Pareto efficient: True.
- Controller signature: nearest_neighbor_multistart:cheapest_insertion_two_opt:edge_preserving_restart:annealed_multi_start:stag4:cand3:restart2.
- Controller rule summary: {"acceptance_bias": 0.01, "candidate_limit_offset": 3, "clustered_heuristic": "cheapest_insertion_two_opt", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Interpretable instance-adaptive selection among a frozen heuristic portfolio with bounded online schedule adjustments driven by stagnation, improvement rate, perturbation failures, and time-budget usage.", "failed_perturbation_threshold": 3, "grid_like_heuristic": "candidate_pruned_two_opt",

"low_improvement_threshold": 0.08, "name": "deterministic_instance_adaptive_frozen_portfolio_v4", "nearest_neighbor_trap_heuristic": "annealed_multi_start", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 2, "stagnation_switch_heuristic": "annealed_multi_start", "stagnation_threshold": 4, "temperature_scale": 1.35, "time_budget_trigger": 0.68, "two_cluster_bottleneck_heuristic": "edge_preserving_restart"}

phase8_llm_evolved_controller_diversity_failure_replay

- Execution mode adaptive_controller with replay diversity_failure.
- Final held-out gap 0.201354, selector regret 0.129094, runtime-adjusted gap 0.201354.
- Family transfer 0.024368 and combined transfer 0.089574.
- Runtime 319.0006 ms, runtime inflation -0.783811, mean novelty 0.702736, and complexity 1.7375.
- Pareto efficient: True.
- Controller signature: nearest_neighbor_multistart:cheapest_insertion_two_opt:farthest_insertion_two_opt:edge_preserving_restart:stag3:cand2:restart2.
- Controller rule summary: {"acceptance_bias": 0.01, "candidate_limit_offset": 2, "clustered_heuristic": "cheapest_insertion_two_opt", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive controller over a frozen TSP heuristic portfolio. Structure-conditioned selection with bounded schedule tuning and an interpretable stagnation/failed-perturbation switch.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.08, "name": "deterministic_instance_adaptive_structure_switch_portfolio_v2", "nearest_neighbor_trap_heuristic": "candidate_pruned_two_opt", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 2, "selection_rules": {"base_by_structure": [{"if": "bottleneck_score >= 0.55 or two_cluster_bottleneck_score >= 0.55", "then": "two_cluster_bottleneck_heuristic"}, {"if": "clusteredness_score >= 0.54 and convex_hull_ratio >= 0.85", "then": "clustered_heuristic"}, {"if": "grid_likeness >= 0.22 and distance_cv >= 0.45", "then": "grid_like_heuristic"}, {"if": "corridor_score >= 0.46 and coordinate_spread_norm >= 0.2", "then": "corridor_heuristic"}, {"if": "nearest_neighbor_trap_score >= 0.30 and nn_distance_cv >= 0.55", "then": "nearest_neighbor_trap_heuristic"}, {"if": "True", "then": "default_heuristic"}], "bounded_schedule_tuning": {"acceptance_bias_rule": "acceptance_bias if recent_improvement_rate <= low_improvement_threshold else 0.0", "candidate_limit_offset_rule": "candidate_limit_offset if (grid_likeness >= 0.22 or corridor_score >= 0.46) else 0", "restart_offset_rule": "restart_offset if stagnation_length >= stagnation_threshold else 0", "temperature_scale_rule": "temperature_scale if (structure_class == 'mixed' and recent_improvement_rate <= low_improvement_threshold) else 1.0"}, "deterministic_priority_hint": ["Diversity_failure mode: if recent_improvement_rate <= low_improvement_threshold and time_budget_used >= time_budget_trigger, prefer random_like_heuristic once; otherwise keep the base rule to avoid thrashing."], "final_resolve_logic": ["step1: choose base via base_by_structure", "step2: if online_stagnation_switch triggers, output stagnation_switch_heuristic", "step3: else if (time_budget_used >= time_budget_trigger and recent_improvement_rate <= low_improvement_threshold) then output random_like_heuristic else keep base", "step4: if time_budget_emphasis overrides, use that result (unless overridden by step2)."], "online_stagnation_switch": [{"if": "stagnation_length >= stagnation_threshold and recent_improvement_rate <= low_improvement_threshold", "then": "stagnation_switch_heuristic"}, {"if": "failed_perturbation_count >= failed_perturbation_threshold and time_budget_used <= time_budget_trigger", "then": "stagnation_switch_heuristic"}, {"if": "True", "then": "base"}], "time_budget_emphasis": [{"if": "time_budget_used >= time_budget_trigger and (grid_likeness >= 0.22 or corridor_score >= 0.46)", "then": "candidate_pruned_two_opt"}, {"if": "time_budget_used >= time_budget_trigger and (bottleneck_score >= 0.55 or two_cluster_bottleneck_score >= 0.55)", "then": "farthest_insertion_two_opt"}, {"if": "time_budget_used >= time_budget_trigger and distance_cv >= 0.55", "then": "nearest_neighbor_multistart"}, {"if": "True", "then": "base"}]}, "stagnation_switch_heuristic": "edge_preserving_restart", "stagnation_threshold": 3, "temperature_scale": 1.15, "time_budget_trigger": 0.6, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}

phase8_full_solver_evolution

- Execution mode full_solver with replay none.
- Final held-out gap 0.125551, selector regret 0.053291, runtime-adjusted gap 0.125551.
- Family transfer 0.001014 and combined transfer 0.046896.
- Runtime 440.231957 ms, runtime inflation -0.701652, mean novelty 0.755521, and complexity 0.76.
- Pareto efficient: True.

Judge Appendix

Conservative Interpretation of Phase 8 TSP Heuristic Portfolio Results

Condition	Final Held-Out TSPLIB Gap	Selector Regret vs Oracle	Runtime-Adjusted Gap	Runtime Inflation	Transfer Gap	Pareto Efficient	Notes Summary
phase8_best_single_fixed_heuristic	**0.07226**	0.0	0.07226	0.0	0.031222	No	Best held-out TSPLIB gap (primary endpoint). Uses static single heuristic (farthest_insertion_two_opt). Zero runtime inflation and selector regret (oracle baseline). No adaptive selection.
phase8_oracle_selector	0.07226	0.0	0.074998	0.037891	0.026678	No	Oracle selector upper bound confirms best fixed heuristic performance on TSPLIB gap. Slight runtime inflation. Not deployable but useful for benchmarking selector regret.
phase8_supervised_ml_selector	0.07226	0.0	0.07226	-0.015596	0.031222	Yes	Static (non-adaptive) learned selector matches best fixed heuristic gap exactly, slightly faster. Interpretable instance-adaptive control is limited but computationally efficient.
phase8_full_solver_evolution	0.125551	0.053291	0.125551	-0.701652	0.046896	Yes	Adaptive full solver evolved controller reduces family gap substantially but TSPLIB performance worse than best fixed. Runtime improved but with some selector regret.
phase8_llm_evolved_controller_diversity_failure_replay	0.201354	0.129094	0.201354	-0.783811	0.089574	Yes	Interpretable adaptive controller with diversity failure replay shows moderate TSPLIB gap and regret, runtime savings, and improves cross-family transfer relative to random baseline.

Condition	Final Held-Out TSPLIB Gap	Selector Regret vs Oracle	Runtime-Adjusted Gap	Runtime Inflation	Transfer Gap	Pareto Efficient	Notes Summary
phase8_llm_evolved_adaptive_controller	0.214311	0.142051	0.214311	-0.773883	0.107596	Yes	Interpretable instance-adaptive controller with bounded online schedule adjustments yields decent runtime tradeoff but noticeably higher gaps and regrets than static selectors.
phase8_llm_static_selector	0.229271	0.157011	0.229271	-0.816997	0.111754	Yes	Deterministic instance-adaptive static selector with bias towards farthest_insertion_two_opt shows highest regret and gap among Pareto points, but substantial runtime savings.
phase8_random_portfolio	0.171446	0.099186	0.171446	-0.745895	0.115633	Yes	Random portfolio selection is Pareto efficient but has clearly worse gaps than best fixed or adaptive selectors, with substantial runtime reduction.

Summary Observations

- **Primary endpoint (held-out TSPLIB gap)** is lowest (best) and tied at ~0.07226 by the **best single fixed heuristic (farthest_insertion_two_opt)**, the **oracle selector** (upper bound), and the **supervised ML static selector**.

- **Selector regret** is zero only for best single fixed heuristic, supervised ML selector, and oracle selector; adaptive controllers incur higher regret up to ~0.14–0.16, indicating suboptimal adaptive control.

- **Runtime inflation** is minimal or negative (improvement over baseline) for all except oracle (slight inflation). Adaptive controllers achieve significant runtime reduction (roughly 70–80%) but at expense of increased TSPLIB gap and regret.

- **Pareto efficiency** includes adaptive controllers and several static selectors, indicating tradeoffs between runtime and solution quality.

- **Cross-family transfer gaps** are lowest (~0.03) for best fixed and supervised ML selectors; adaptive controllers have moderately higher transfer gaps (~0.09-0.11) but still better than random portfolio (~0.11).

- Adaptive controllers provide **interpretable instance-adaptive control** with structured heuristics switching and schedule tuning. However, they do **not approach the oracle or best fixed heuristic TSPLIB gap**, and improvements come primarily as runtime savings rather than solution quality.

- No evidence of new algorithmic heuristics; results confirm effective **hyper-heuristic control over known heuristics**.

Conservative Conclusion and Recommendations

1. The **best single fixed heuristic (farthest_insertion_two_opt)** remains the strongest baseline on held-out TSPLIB gap without runtime overhead.

2. The **oracle selector** validates the ceiling performance; static selectors that approach oracle gap without sizable runtime inflation (e.g., supervised ML selector) offer promising interpretable instance-adaptive control.

3. Adaptive controllers yield meaningful **runtime advantages with moderate quality tradeoffs**, indicating practical value when fast approximate solutions are acceptable with transparent control policies.

4. Further tuning to reduce selector regret and TSPLIB gap in adaptive controllers is needed before matching best fixed heuristic.

5. Random portfolio strategy is inferior both in gap and regret compared to learned or adaptive selectors, confirming benefit of adaptive hyper-heuristics.

Overall, the phase 8 evaluation confirms that instance-adaptive hyper-heuristic control can effectively trade off between solution quality and runtime, but the **best fixed heuristic remains a strong, simple baseline for held-out TSPLIB performance**. Interpretability and bounded adaptation in evolved controllers show good promise but require refinement to approach oracle-like solution quality.